

Systems

Under Pressure

Investigation Notebook

DRIVING QUESTION

How do we know when a system is actually in danger?

PROGRAM TOPIC



NOTEBOOK STRUCTURE

Pages 1-3 · Empathy + Explore
Pages 4-6 · Systems + Intervals
Pages 7-9 · Build Models
Pages 10-12 · Critique + Revise
Pages 13-15 · Claim + Present

HOW TO USE THIS NOTEBOOK

This is a field journal, not a worksheet.
Write, sketch, map, and calculate anywhere on the page.
Show your thinking, including what changed.
Claim -> Evidence -> Test -> Revision
Vlogs are formative checkpoints (4-6 min each): record after each class and upload with your individual artifact and written reflection.
Written reflection: minimum 150 words per vlog.

Start here before categories, before models, before data. Understand the situation first. Empathy informs which variables and thresholds are worth investigating.

"Who are the people most affected by this type of system and what do they experience when it fails?"

"Who benefits when this system is healthy? Who suffers most and first when it fails, and why?"

"What does someone who disagrees with the mainstream view argue, and what evidence do they use?"

Before building models, understand who is affected and how. Use AI to gather perspectives, then decide which variables matter.

"Generate stakeholder perspectives on the topic, including conflicting interests."

Variables + Hidden Systems

Map connected variables and systems. Think in pressures, delays, and ripple effects. Name at least one confounding variable.

AI CO-CREATION PROMPTS

"What variables are connected to your topic that are rarely measured?"

"If variable A increases, what else might change, and with what time delay?"

"What confounding variable would a skeptic point to if I claimed A causes B?"

CONNECTED SYSTEMS MAP - DRAW ARROWS, PRESSURES, CONNECTIONS

VARIABLES WE CAN MEASURE

VARIABLES WE CANNOT EASILY MEASURE

REQUIRED SENTENCE FRAME: These variables appear related because ____, but this does not prove causation because ____.

For each source record: variable measured, timeframe, credibility, limitations, and keep or reject.

Slow down. Annotate before interpreting. Name thresholds, turning points, anomalies, and what the graph does not show.

Local behavior is a short interval. Global behavior is the long-term pattern. Choosing an interval is a mathematical decision.

Sequences & Series Modeling

Investigate repeated change. Build a ratio/difference table. Use plain notation: a_n , $a_{(n-1)}$, S_n , and $r^{(n-1)}$.

n	Value a_n	Difference $a_n - a_{(n-1)}$	Ratio $a_n / a_{(n-1)}$	Ratio stable?	Notes
ARITHMETIC OR GEOMETRIC? JUSTIFY.		RECURSIVE: $a_n = r \cdot a_{(n-1)} + d$		EXPLICIT: $a_n = a_1 \cdot r^{(n-1)}$	

PARTIAL SUM: $S_n = a_1(1-r^n)/(1-r)$

WHERE DOES THE PATTERN BREAK?

Exponentials + Thresholds

Test whether a common ratio is stable enough for $f(x)=a*b^x$. Solve thresholds with $\log_b(threshold/a)=x$.

x	y value	y / y_previous	Ratio stable?	Notes
EXPONENTIAL MODEL: $f(x)=a*b^x$ - WHAT DO a AND b MEAN?			CONTINUOUS $f(x)=a*b^x$ OR DISCRETE $a_n=a_1*r^{(n-1)}$?	

THRESHOLD SOLVE: $\log_b(threshold/a)=x$

WHERE DOES EXPONENTIAL REASONING FAIL?

Focus on graph shape over a local interval. Identify turning points, zeros, multiplicity, and real thresholds.

WHAT DOES $P(t)$ REVEAL THAT $f(x)=a \cdot b^x$ OR A LINEAR MODEL CANNOT?

Model Comparison

Compare all model types for the same data. Each reveals something. Each hides something.

CRITERION	Linear	Geometric / Exponential	Polynomial P(t)	Arithmetic Sequence

MOST DEFENSIBLE MODEL - AND WHY THE OTHERS ARE LESS APPROPRIATE

Confounding Variables

A confounding variable is a third variable that could explain the relationship between your two main variables.

CONFOUNDING VARIABLE 1 - NAME + EFFECT

CONFOUNDING VARIABLE 2 - NAME + EFFECT

CONFOUNDING VARIABLE 3 - NAME + EFFECT

CONFOUNDING VARIABLE 4 - NAME + EFFECT

REQUIRED SENTENCE FRAME: These variables appear related because ____, but this does not prove causation because ____.

Document meaningful use of Flint. Protocol: Think First -> Flint -> Verify -> Decide.

AI CO-CREATION PROMPTS

ENTRY 2: prompt, suggestion, accepted/rejected, verification

ENTRY 3: prompt, suggestion, accepted/rejected, verification

Revision Trail

Make changing thinking visible. Every significant revision should be traceable to evidence or contradiction.

ORIGINAL CLAIM

APPROPRIATE PROMPTS

EVIDENCE THAT CONTRADICTED IT

REVISED INTERPRETATION - WHAT CHANGED, WHY, AND WHAT EVIDENCE FORCED IT

REMAINING UNCERTAINTY

What is your most responsible mathematical interpretation? Name your model, evidence, limitation, and uncertainty.

Plan your mathematical argument, not a summary of what you did. Know your strongest evidence and weakest assumption.